

Segment 4 → Greedy Method

Greedy strategy to randomly pick the best option at each step.

■ Differences between Dynamic Programming & Greedy Algorithm:

Dynamic Programming	Greedy Algorithm
At each step, the choice is determined based on solutions of subproblems.	At each step, we quickly make a choice that currently looks best.
Sub-problems are solved first.	Greedy choice can be made first before solving further subproblems.
Bottom-up approach	Top-down approach.
Can be slower, more complex.	Usually faster, simpler.

* Activity Selection Problem *

6 marks Go on
Exam exam-2

■ The problem is to select a maximum-size set of mutually compatible activities is called activity selection problem.

■ $s_i \rightarrow$ Start time

$f_i \rightarrow$ Finish time

Consider the following set S of activities:

i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14

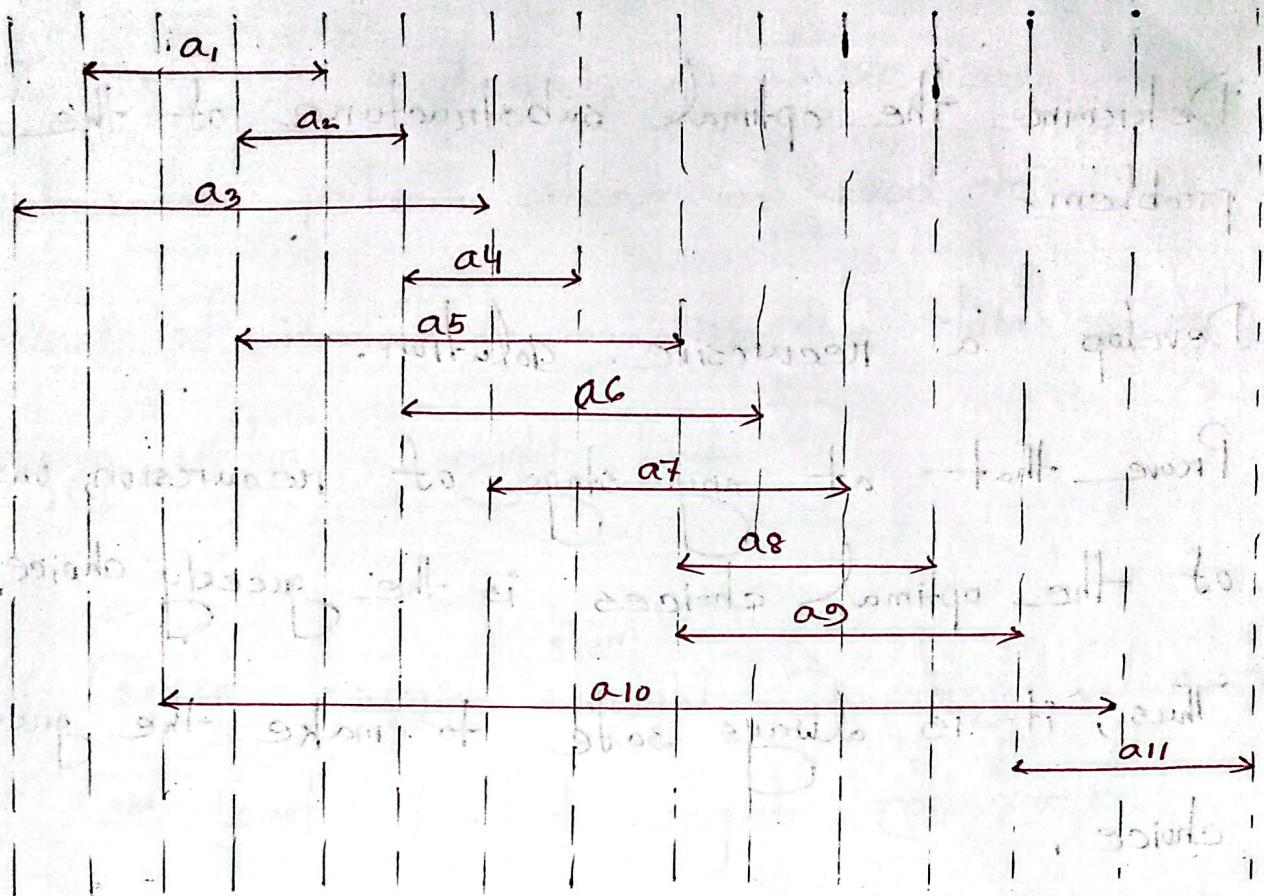
* Mutually compatible activities $\rightarrow$ Greedy sequence

ଆହୁରି sequence ଟା overlap

କରା ନା ।

* ଗ୍ରେଡିଆନ୍ ଅର୍ଥାତ୍ ଅନ୍ୟକ sequence ହୁଏନା ଓ ନିୟତ ହେଉ ଟପାଆ
ଟପାଆଟି ଆହୁରିକ overlap କରା ନା ।

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14



All possible mutually compatible activities are:

$\{a_1, a_4, a_8, a_{11}\}$, $\{a_3, a_7, a_{11}\}$, $\{a_2, a_4, a_8, a_{11}\}$,

$\{a_2, a_4, a_9, a_{11}\}$

Largest set of mutually compatible activities are:

$\{a_1, a_4, a_8, a_{11}\}$, $\{a_2, a_4, a_8, a_{11}\}$ and $\{a_2, a_4, a_9, a_{11}\}$

* Steps of the Greedy Strategy *

→ Theory ଓ
ଆଲୋଚନା ଆଲୋଚନା

1. Determine the optimal substructure of the problem.
2. Develop a recursive solution.
3. Prove that at any stage of recursion, one of the optimal choices is the greedy choice, Thus; it is always safe to make the greedy choice.
4. Show that all but one of the subproblems induced by having made the greedy choice are empty.
5. Develop a recursive algorithm that implements the greedy strategy.
6. Convert the recursive algorithm to an iterative algorithm.

* Knapsack Problem *

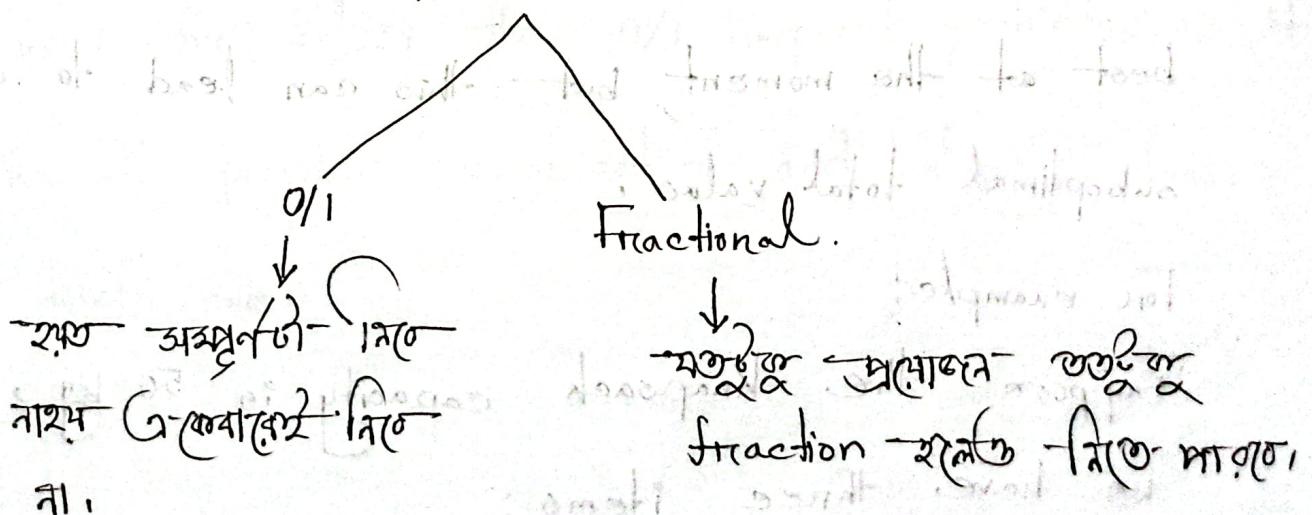
■ The knapsack problem is a classic combinatorial optimization problem where we need to choose a subset of items to maximize the total value while staying within a weight limit.

અર્થઃ આપણે ^{item} વજન (weight) એક, અને value હદ
શરૂ થઈ, અને limited weight હદા શરૂ.

અતઃ item select કરી શકીએ limited weight

કે- અર્થઃ અલગ value માત્રા સાથે.

Knapsack Problem



* Knapsack 0/1 can not be solved by greedy strategy — Ex - Given 3 marks Q - Ans -

⇒ The 0/1 knapsack problem can not be solved by a greedy strategy because greedy algorithms make decisions based only on immediate benefits such as the highest value or the best value-to-weight ratio without considering the overall best combination of items.

In the 0/1 knapsack problem, each item can either be taken entirely or not taken at all. The greedy approach may choose items that seem best at the moment, but this can lead to a suboptimal total value.

For example:

Suppose the knapsack capacity is 50 kg, and we have three items:

Item A: Value = 60, weight = 10

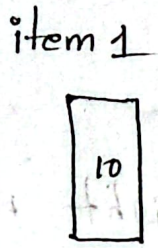
Item B: value = 100, weight = 20

Item C: value = 120, weight = 30

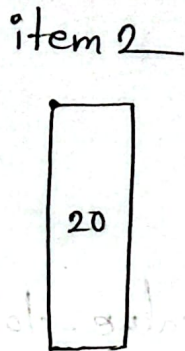
The greedy approach based on value-to-weight ratio would choose A(6), then B(5). for a total weight of 30 kg and total value of 160. However, the better choice is B(5) and C(4), which fit exactly into 50 kg and give a higher total value of 220.

Therefore, the greedy strategy fails to guarantee the optimal solution. That's why dynamic programming is used for solving the 0/1 Knapsack problem, as it checks all possible combinations and always finds the best one.

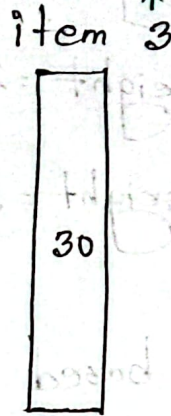
* Fractional Knapsack *



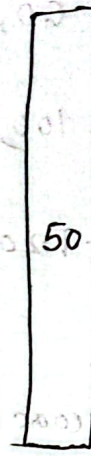
\$60



\$100



\$120



Knapsack

এটা solve করতে কালে সমস্যাটা না নিয়ে লোভানী থেকে
মতুইক প্রয়োজন ততুইকুও নেয়া যাবে।

⇒ Taking the items in order of greatest value
per pound yields an optimal solution

$\frac{20}{30}$
20
10

\$80

\$100

\$60

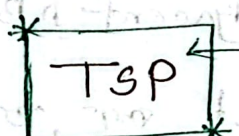
Total = \$240

આમલે item 2 ના, item 3 ની 50 fillup થઈ (મત
ફિલ્ડ) અને value થઈ 220.

આમલે item 3 અમુલ ની, item 1 અમુલ ની
item 2 થઈ 20 ની value થઈ 230, જે
best ની.

આમલે pick કરીને highest value (per weight)

ફૂલો અમુલ / ની ની થઈ.

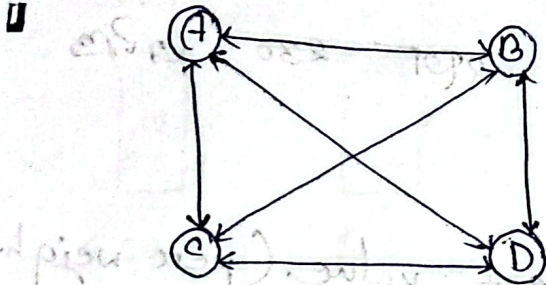
*  * Travelling Salesman Problem

The travelling salesman problem consists of a
salesman and a set of cities.

The salesman has to visit each one of the
cities starting from a certain one and returning
to the same city.

The challenge of the problem is that the
travelling salesman wants to minimize the total
length of the trip.

এই problem-কে Greedy method-এ ও Dynamic method-এ উল্লিখিত করে।



	A	B	C	D
A	0	10	15	20
B	5	0	25	10
C	15	30	0	5
D	15	10	20	0

① Greedy Method:

এখানে সবার মধ্যে সবচেয়ে কম cost-এর edge-কে (Adjacent D, B) নিয়ে নেওয়া হবে।

এখানে C থেকে D-এর cost-এর চেয়ে D-এর cost-এর চেয়ে কম। তাই D-কে নেওয়া হবে।

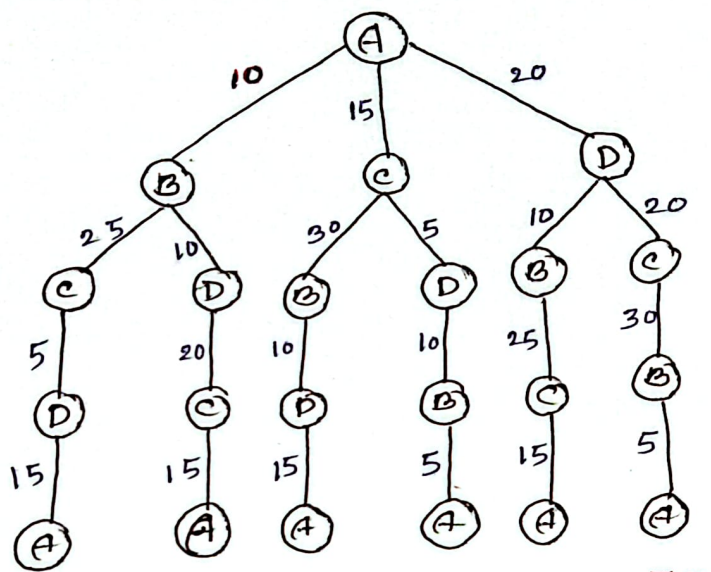
D-এর A, B-তে আর back করতে হবে না। এতে পারে, C-এর A-তে back করতে হবে।

So, path হবে $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$

তার total cost হবে $10 + 10 + 20 + 15 = 55$

② Dynamic Method:

ଓଢ଼ି method ଓ ଅସମ୍ଭାବ୍ୟ possible path check କର ଅସମ୍ଭାବ୍ୟ
କାସ cost ଓଢ଼ି ଓଢ଼ି ସାହାଯ୍ୟ କରୁଅଛି ।



Total cost $\Rightarrow$

55

55

70

35

70

75

ଓଢ଼ି² min cost.

ଓଢ଼ି process-ଓ ଓଢ଼ି କାସକାସ କର ଅସମ୍ଭାବ୍ୟ possible
path check କର ଅସମ୍ଭାବ୍ୟ । ଓଢ଼ି time complexity ଓଢ଼ି
ଓଢ଼ି ସାହାଯ୍ୟ ।